

Mixed-Frame Building — Comprehensive Structural Report

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Project Overview

Project Description: 21m Span Industrial Building **Configuration:** 6 Frames total (5 Standard Frames + 1 Long-Span Frame) **Standard Frame:** 21m Isosceles Howe Truss with 2 interior RC columns. **Long-Span Frame:** 21m Isosceles Howe Truss with NO interior columns.

Part A: Standard Frame Structural Analysis

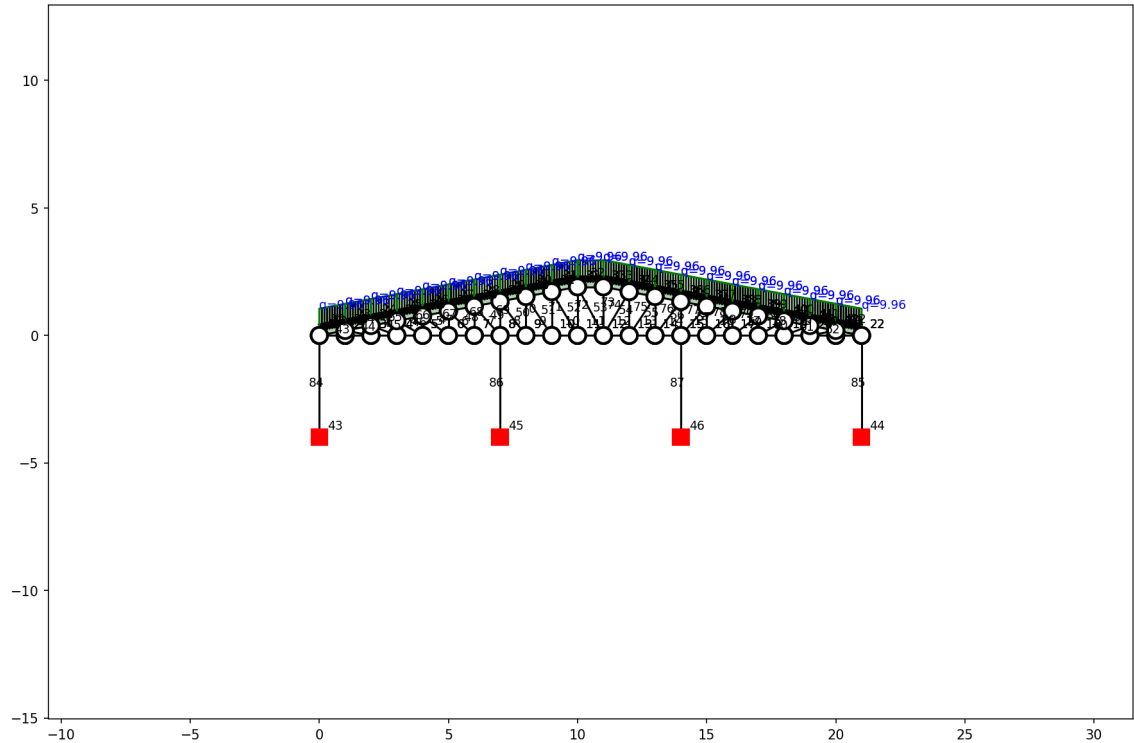
Derived from Bar_Project_RC_Final

1. Truss Configuration

Parameter	Value
Span	21.0 m
End Depth	0.0 m (Isosceles Triangle)
Depth at Midspan (Peak)	2.0 m
Bottom Chord	Level at y = 0.0 m
Column Height	4.0 m (Fixed base, RC columns)
Panels	21 @ 1.0 m
Type	Isosceles Triangle Howe Truss

Chord Construction	Double Angles (2L) — back-to-back for gusset plates
Web Construction	Square or Rectangular Pipe (Cold Rolled Tubular Steel)
Connections	Site-welded using 6mm or 10mm gusset plates
Supports	4 fixed-base RC columns: 2 exterior + 2 interior
Interior Column Locations	Exactly at 7.0 m and 14.0 m
Steel Grade	A36 (Fy = 36 ksi)
Concrete Grade	21 MPa (3000 psi)

Standard Frame Truss Diagram



2. Applied Loads

Gravity Loads

Component	Value
Dead Load (Roof + Ceiling)	3.0 kN/m (0.5 kPa x 6m trib.)
Truss + RC Col Self-Weight	3.057 kN/m
Total Dead Load (D)	3.500 kN/m
Live Load (L)	3.600 kN/m (0.6 kPa x 6m trib.)

Earthquake (Static Method — NSCP)

Parameter	Value
Seismic Coefficient (Cs)	0.2 (Zone 4, Soil Type D, R=8.5)
Seismic Weight (W)	73.50 kN
Base Shear (V = Cs x W)	14.70 kN
Lateral q per exterior column (V / 4 / h)	0.92 kN/m

Wind Load

Parameter	Value
Wind Pressure	800 Pa
Tributary Width	6.0 m
Lateral Load on Ext. Columns	4.8 kN/m (800 Pa x 6m)

3. Load Case Summary

Axial Forces (kN) per Load Case

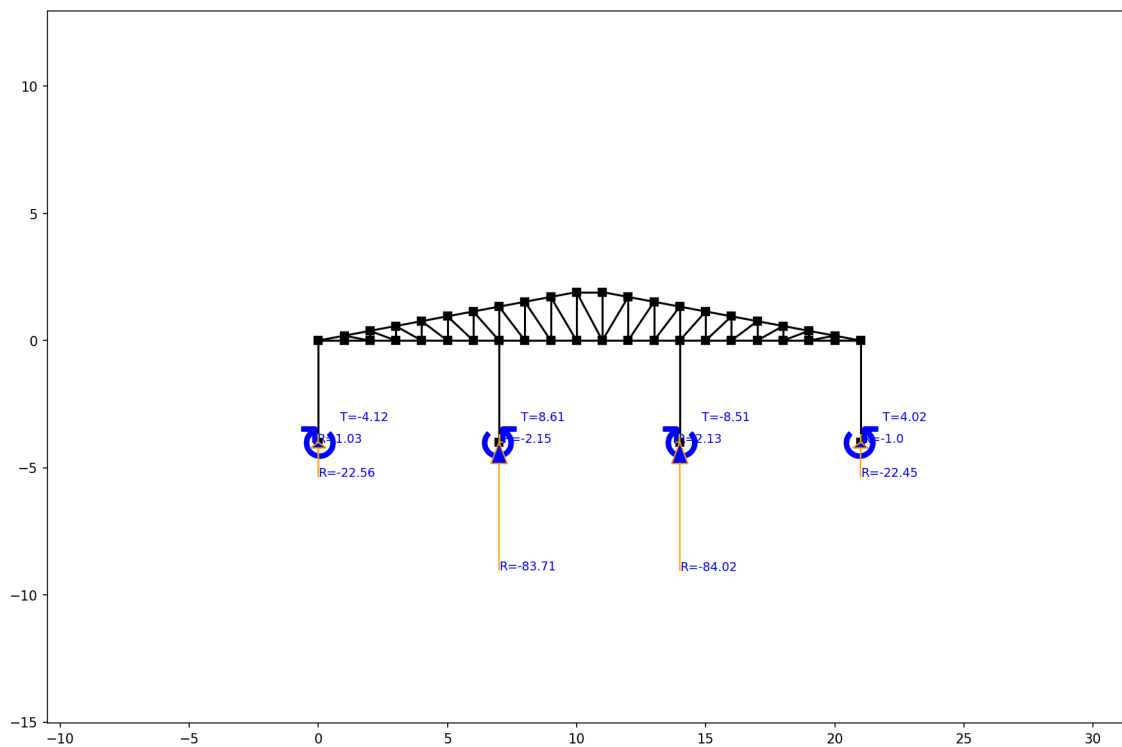
Member Group	LC1 — Gravity Only	LC2 — Gravity + EQ	LC3 — Gravity + Wind	Governing
Top Chord	-94.41	-74.16	-40.95	-94.41
Bottom Chord	90.78	71.14	40.42	90.78
Vertical Webs	-53.60	-42.10	-23.24	-53.60
Diagonal Webs	47.54	37.36	20.75	47.54
Exterior Columns	22.56	17.71	9.73	22.56
Interior Columns	84.02	65.92	36.07	84.02

Axial Forces Diagram (Standard Frame)



Governing Vertical Reaction (Exterior): 22.56 kN Governing Vertical Reaction (Interior): 84.02 kN Governing Horizontal Reaction: 16.15 kN

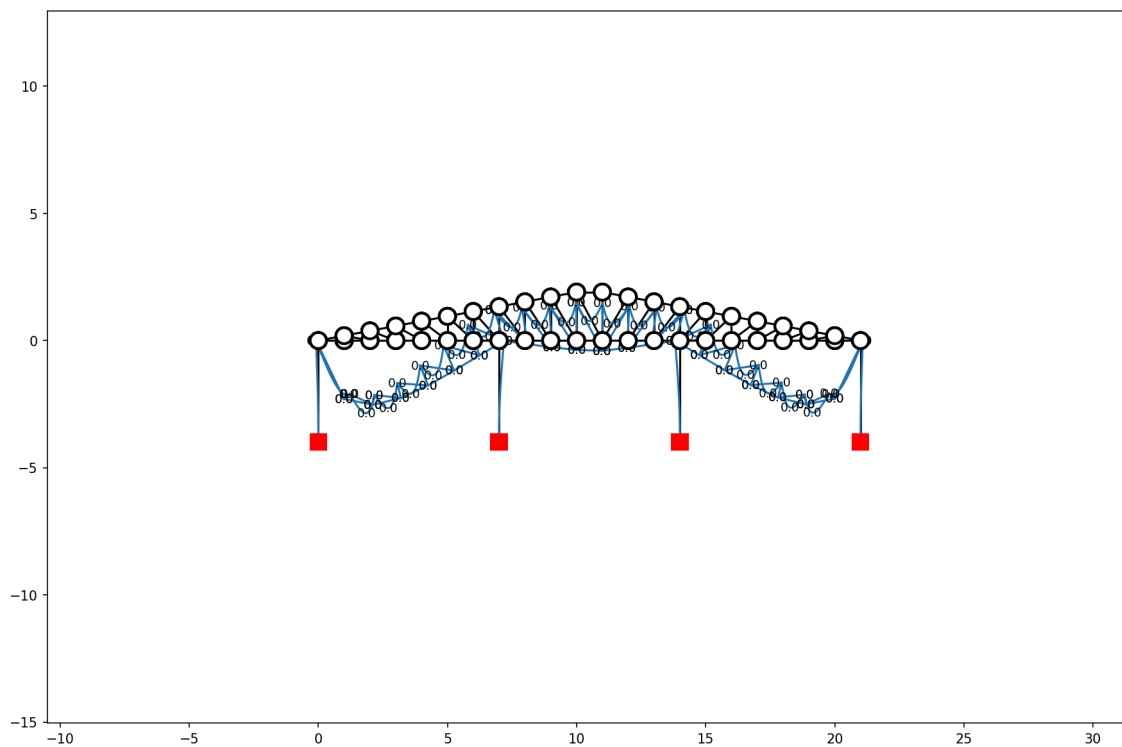
Support Reactions Diagram (Standard Frame)



4. Deflection Check

- **Max Deflection:** 0.01 mm
- **Deflection Ratio (L/d):** 1999505 (Target > 240) PASS

Deflection Diagram (Standard Frame)



5. Optimized Member Selection (Governing Forces)

Group	Selected Shape (Standard / Comm.)	Weight	KL/r	Gov. Force (kN)	Capacity (kN)	Governing LC
Exterior Columns	400x400 RC	384.0 kg/m	0.0	22.56	0.00	Gravity Only
Interior Columns	400x400 RC	384.0 kg/m	0.0	84.02	0.00	Gravity Only
Top Chord	2L2X2X1/8	3.3 plf	64.6	-94.41	98.24	Gravity Only
Bottom Chord	2L2X2X1/8	3.3 plf	63.5	90.78	141.53	Gravity Only
Vertical Webs	HSS2X2X1/8 / Square Pipe 51x51x3.2mm	3.0 plf	103.5	-53.60	68.90	Gravity Only
Diagonal Webs	HSS2X1X1/8 / Rect. Pipe 51x25x3.2mm	2.2 plf	225.7	47.54	87.63	Gravity Only

- Total Steel Weight: 401.0 kg
- Total Concrete Weight (Cols): 6144.0 kg

6. Substructure Design (Standard Frame)

Exterior Column Foundation (x2)

Column: 400x400 RC, Height: 4.00 m, Pu = 22.56 kN

Parameter	Value
RC Column Section	400 mm x 400 mm
Vertical Reinforcement	8 — 16mm bars (Grade 40)
Lateral Ties	10mm @ 200mm O.C.
Isolated Footing	1.0 m x 1.0 m x 0.40 m
Bottom Rebar	12mm @ 200mm O.C. (Both Ways)

Interior Column Foundation (x2)

Column: 400x400 RC, Height: 4.00 m, Pu = 84.02 kN

Parameter	Value
RC Column Section	400 mm x 400 mm
Vertical Reinforcement	8 — 16mm bars (Grade 40)
Lateral Ties	10mm @ 200mm O.C.
Isolated Footing	1.0 m x 1.0 m x 0.40 m
Bottom Rebar	12mm @ 200mm O.C. (Both Ways)

Part B: Long-Span Frame Structural Analysis

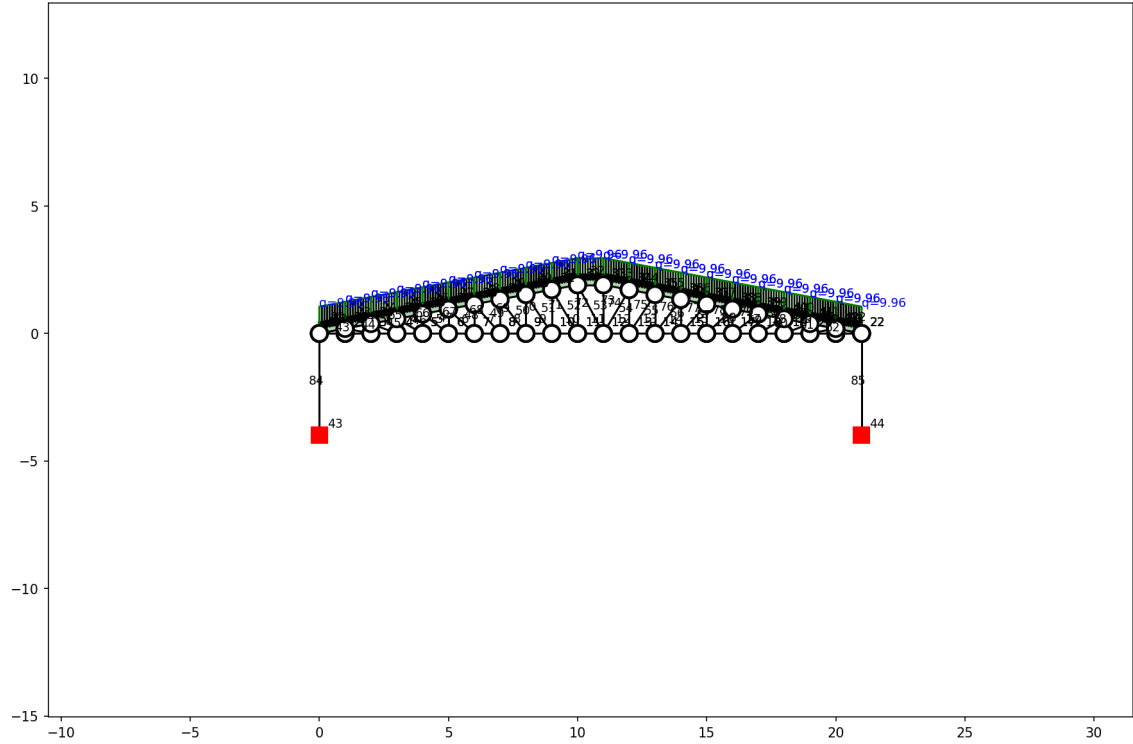
Derived from Bar_Project_Long_Span

1. Truss Configuration (Long Span)

Parameter	Value
Span	21.0 m
End Depth	0.0 m (Isosceles Triangle)
Depth at Midspan (Peak)	2.0 m
Bottom Chord	Level at y = 0.0 m
Column Height	4.0 m (Fixed base, RC columns)
Panels	21 @ 1.0 m
Type	Isosceles Triangle Howe Truss

Chord Construction	Double Angles (2L) — back-to-back for gusset plates
Web Construction	Square or Rectangular Pipe (Cold Rolled Tubular Steel)
Connections	Site-welded using 6mm or 10mm gusset plates
Supports	2 fixed-base RC columns: Exterior only (Long Span)
Steel Grade	A36 (Fy = 36 ksi)

Long-Span Frame Truss Diagram

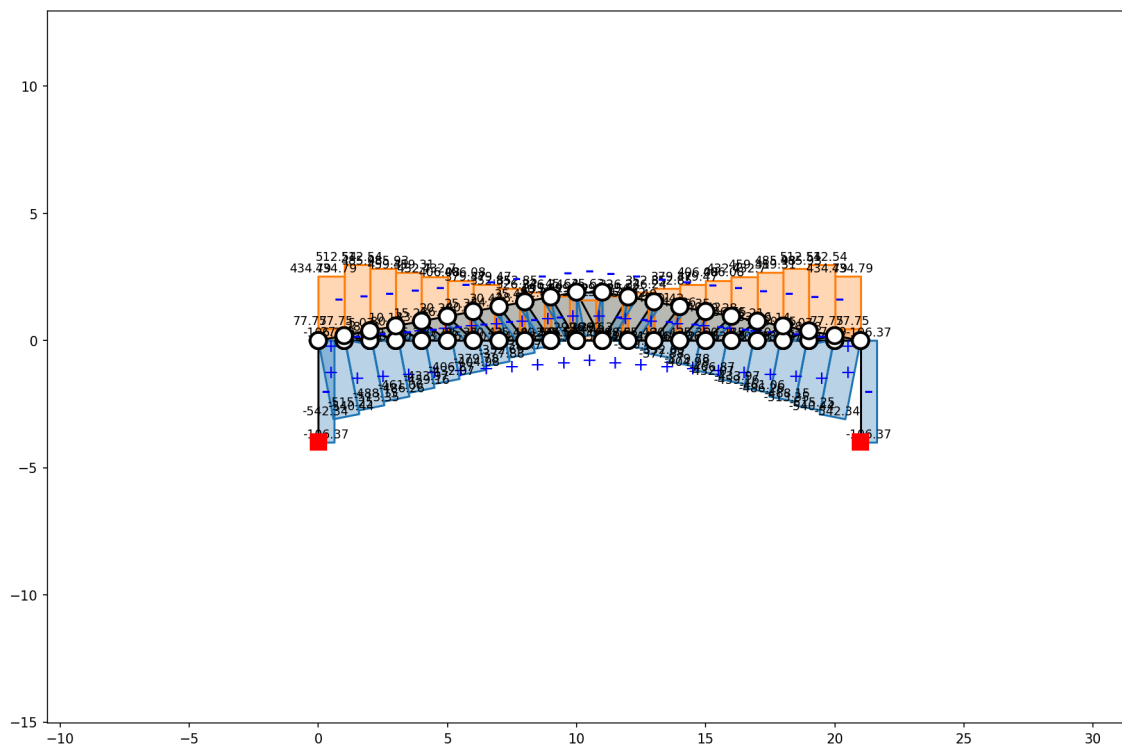


2. Load Case Summary (Long Span)

Axial Forces (kN) per Load Case

Member Group	LC1 — Gravity Only	LC2 — Gravity + EQ	LC3 — Gravity + Wind	Governing
Top Chord	-542.34	-424.73	-228.70	-542.34
Bottom Chord	512.54	401.39	216.13	512.54
Vertical Webs	45.63	35.73	19.24	45.63
Diagonal Webs	77.75	60.89	32.79	77.75
Exterior Columns	106.37	83.30	44.86	106.37

Axial Forces Diagram (Long-Span Frame)

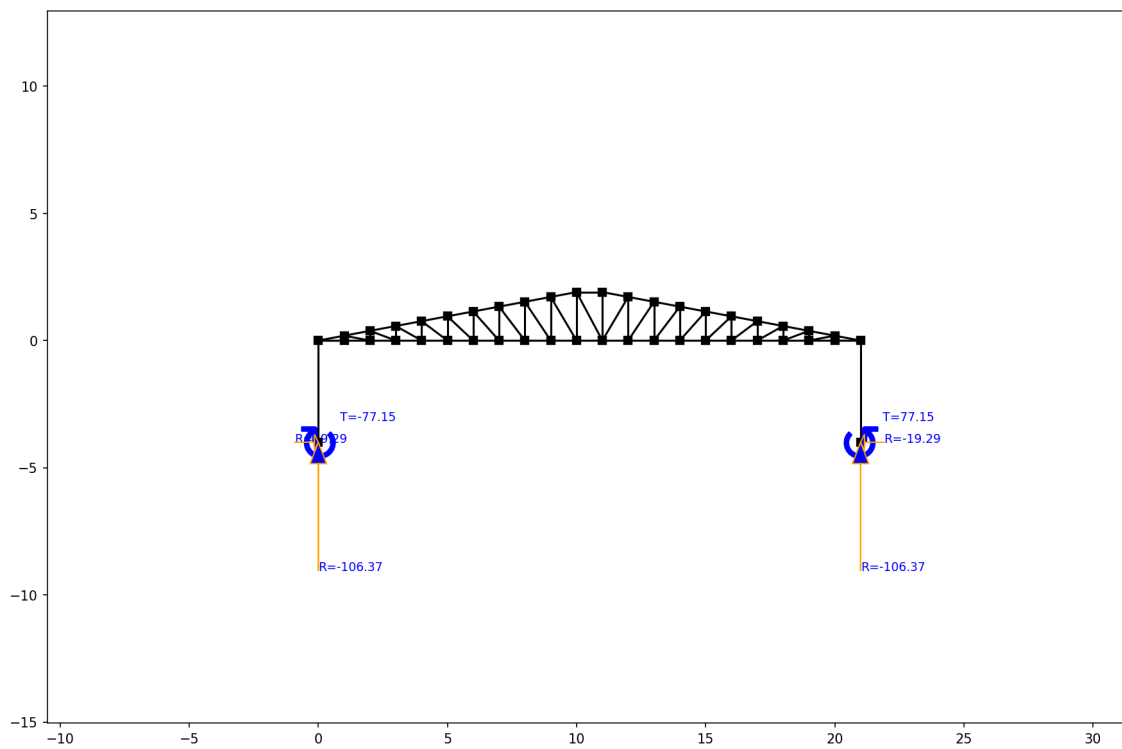


Support Reactions (Long Span)

Reaction	LC1 — Gravity Only	LC2 — Gravity + EQ	LC3 — Gravity + Wind	Governing
Vertical (Fy) — Max	106.37	83.30	44.86	106.37
Vertical (Fy) — Exterior	106.37	83.30	44.86	106.37

Governing Vertical Reaction (Exterior): 106.37 kN Governing Horizontal Reaction: 27.33 kN

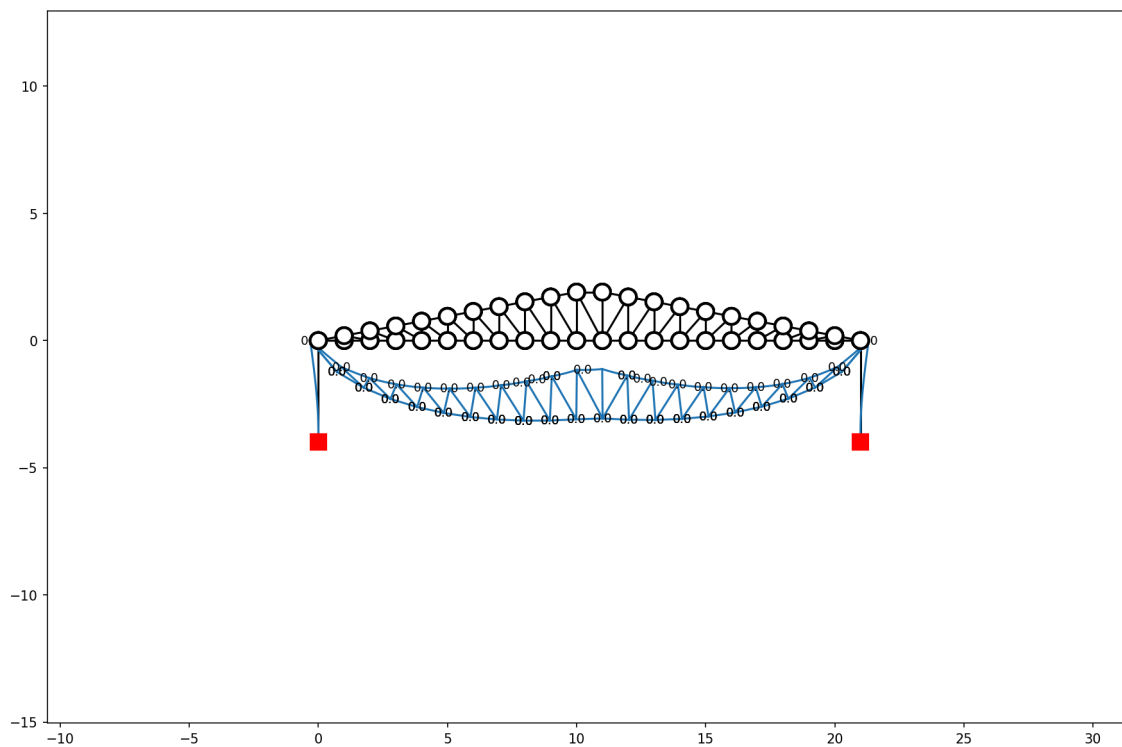
Support Reactions Diagram (Long-Span Frame)



3. Deflection Check (Long Span)

- **Max Deflection:** 0.07 mm
- **Deflection Ratio (L/d):** 302814 (Target > 240) PASS

Deflection Diagram (Long-Span Frame)



4. Optimized Member Selection (Long Span)

Group	Selected Shape (Standard / Comm.)	Weight	KL/r	Gov. Force (kN)	Capacity (kN)	Governing LC
Exterior Columns	400x400 RC	384.0 kg/m	0.0	106.37	0.00	Gravity Only
Top Chord	2L3X3X3/8	14.4 plf	44.0	-542.34	549.16	Gravity Only
Bottom Chord	2L3-1/2X3-1/2X1/4	11.6 plf	36.1	512.54	490.02	Gravity Only
Vertical Webs	HSS2X1X1/8 / Rect. Pipe 51x25x3.2mm	2.2 plf	201.9	45.63	87.63	Gravity Only
Diagonal Webs	HSS2X1X1/8 / Rect. Pipe 51x25x3.2mm	2.2 plf	225.7	77.75	87.63	Gravity Only

- Total Steel Weight: 986.7 kg
- Total Concrete Weight (Cols): 3072.0 kg

5. Substructure Design (Long Span)

Exterior Column Foundation (x2)

Column: 400x400 RC, Height: 4.00 m, $P_u = 106.37$ kN

Parameter	Value
RC Column Section	400 mm x 400 mm
Vertical Reinforcement	8 — 16mm bars (Grade 40)
Lateral Ties	10mm @ 200mm O.C.
Isolated Footing	1.0 m x 1.0 m x 0.40 m
Bottom Rebar	12mm @ 200mm O.C. (Both Ways)

Tie Beam Details

- Size: 300mm x 400mm RC Tie Beam
- Reinforcement: 4 - 16mm main bars with 10mm @ 200mm stirrups
- Purpose: Seismic connectivity between footings along building length.

Part C: Consolidated Bill of Materials

Derived from Bar_Project_Mixed

Project: Bar building with 5 Standard Frames and 1 Long-Span Frame **Configuration:** 6 Frames total, 21m Span, 4m Height.

- Standard Frames (1-5): Isosceles Howe with 2 interior columns.
- Long-Span Frame (6): Isosceles Howe with NO interior columns.
- Tie Beams: 300x400mm RC beams connecting all exterior footings (126m total length).

1. Summary of Quantities

Material Type	Unit	5x Standard	1x Long-Span	Tie Beams	Grand Total
Structural Steel	kg	2005.0	986.7	-	2991.7
Concrete (21 MPa)	m3	20.80	2.08	15.12	38.00
Reinforcement	kg	1468.5	146.9	1343.2	2958.6

2. Discussion

- Frame 6 (Long Span) requires significantly heavier steel sections (2L3x3x3/8 vs 2L2x2x1/8) to bridge 21m without interior columns.
- Tie beams (300x400) connect all exterior footings along the building length (21m per bay/frame).
- All designs maintain L/240 deflection at service loads.